

Chapter 41. Complications Associated with Mechanical Ventilation

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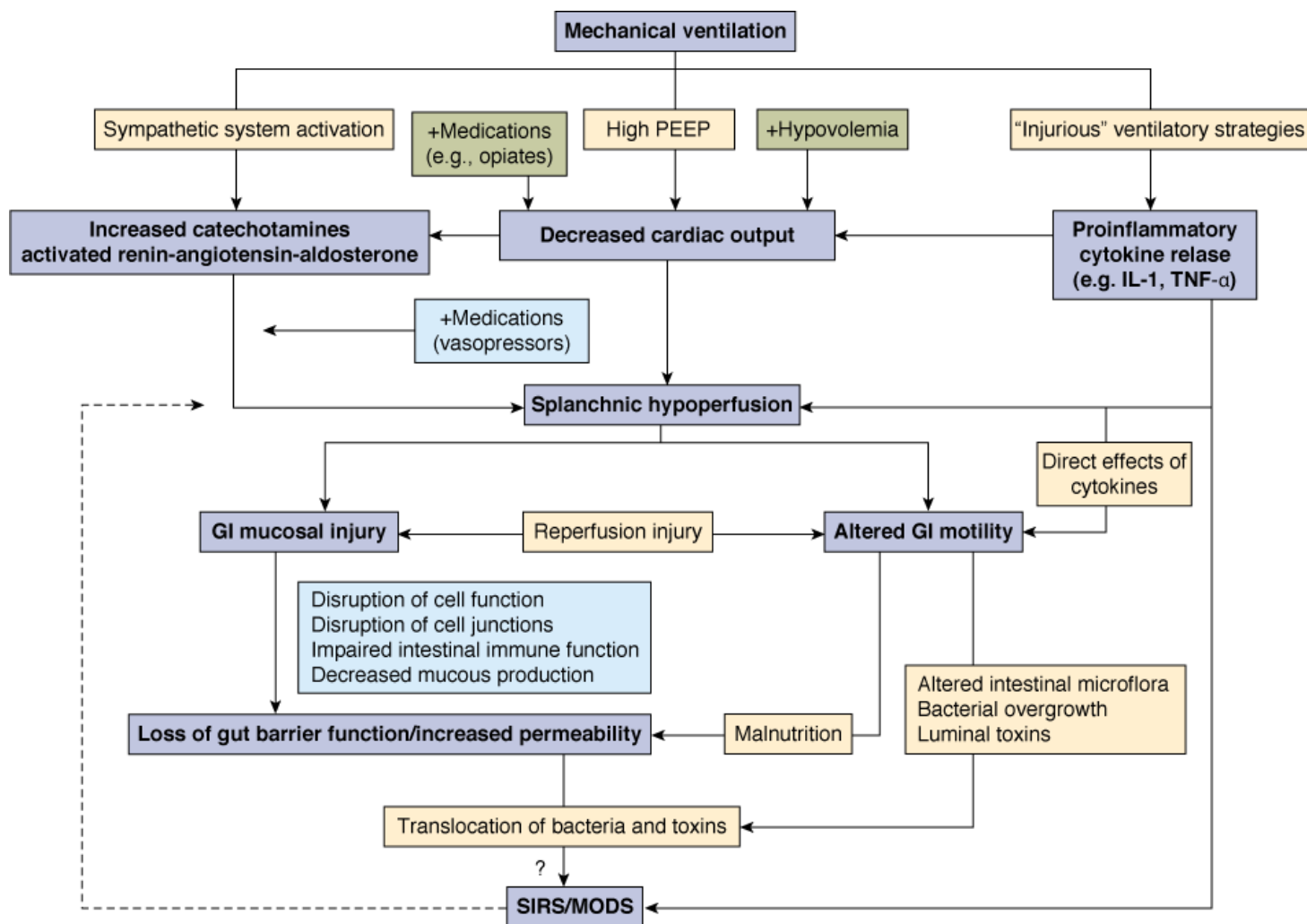
Complications Associated with Mechanical Ventilation: Introduction

Critically ill patients are at risk of succumbing to their primary disease or the undesired sequelae associated with their therapy. Although frequently lifesaving, the use of positive airway pressure therapy has numerous undesired physiologic and clinical complications. These complications have their origins in the endotracheal or tracheostomy tube, the positive-pressure ventilation (PPV), or from therapies delivered during the care of mechanical ventilation. Other complications may result from coexisting illness or comorbid conditions. Although not commonly recognized as important effects of PPV or positive end-expiratory pressure (PEEP), alterations in organ functions ought to be recognized and addressed accordingly to reduce morbidity and mortality.

Gastrointestinal Tract Complications

The interaction between PPV and the gastrointestinal tract in critical setting is a complex one. Gastrointestinal changes are reported frequently in critically ill patients receiving PPV. The true incidence of gastrointestinal complications is not known, but it is reported to be up to 100% for those receiving PPV for more than 3 days. Splanchnic hypoperfusion seems to play a pivotal role in the pathogenesis of these complications, including mucosal damage, motility disorders, and mesenteric ischemia (Fig. 41-1).¹ Unlike many other vascular beds, the splanchnic exhibits limited autoregulation when faced with reduction in mean arterial pressure.² Without altering total blood flow to the organs of the digestive tract, sympathetic stimulation redistributes blood flow to the muscularis of the wall by decreasing mucosal perfusion.³ This poses severe ischemic changes to the mucosal layer because the mucosa is metabolically more active than the muscle layer and is more vulnerable to the destructive effects of a compromised blood supply.

Figure 41-1



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Suggested mechanisms for the development of gastrointestinal (GI) complications during mechanical ventilation. *IL*, interleukin; *MODS*, multiple-organ dysfunction syndrome; *PEEP*, positive end-expiratory pressure; *SIRS*, systemic inflammatory response syndrome; *TNF*, tumor necrosis factor. (Used, with permission, from Mutlu.¹)

Mechanical ventilation influences the gastrointestinal function by impacting systemic hemodynamics via high PEEP or potentially injurious ventilator strategies such as a high tidal volume (V_T). The effect of PEEP on splanchnic blood flow has been shown in animal models to be dose-dependent.⁴ PEEP decreases venous return and reduces preload, which in turn reduces cardiac output and results in splanchnic hypoperfusion.⁵ The reduction of splanchnic blood flow is limited at PEEP levels below 10 cm H₂O but it is more pronounced at PEEP levels of 15 to 20 cm H₂O.⁶ In rats, the addition of 10 cm H₂O of PEEP resulted in reduction of cardiac output and mesenteric blood flow by 31% and 75%, respectively.⁷ PEEP also promotes plasma-renin-angiotensin-aldosterone activity, as well as catecholamine release, which limits splanchnic hypoperfusion.^{4,7} Interestingly, high PEEP levels interfere with mesenteric leukocyte–endothelial interaction. In rats with healthy lungs, 10 mbar of PEEP was associated with an increase in the number of rolling, adherent, and migrated leukocytes when compared with anesthesia alone or with mechanical ventilation with 0 or 5 mbar PEEP.⁸

Alternatively, mechanical ventilation with high V_T can modify the inflammatory responses irrespective of the underlying lung injury. Experimental data suggest that mechanical ventilation with high V_T and zero end-expiratory pressure induces not only cytokine release (tumor necrosis factor [TNF]- α and interleukin [IL]-8) but also translocation of cytokines from the lungs to the systemic circulation and vice versa.^{9,10} These cytokines lead to a number of clinical sequelae in the gastrointestinal tract including splanchnic hypoperfusion and intestinal smooth muscle impairment.^{11,12}

Indirectly, medications administered to patients on mechanical ventilation can have deleterious effects on gastrointestinal function. Opiates and sedatives, such as benzodiazepines, can decrease gut motility and impair venous return.¹³ Other agents, like vasopressors or inotropes,^{14,15} may alter hemodynamic parameters that, in turn, reduce mesenteric blood flow and put a critically ill patient at risk of developing stress-related mucosal disease.⁴

Stress-Related Mucosal Disease

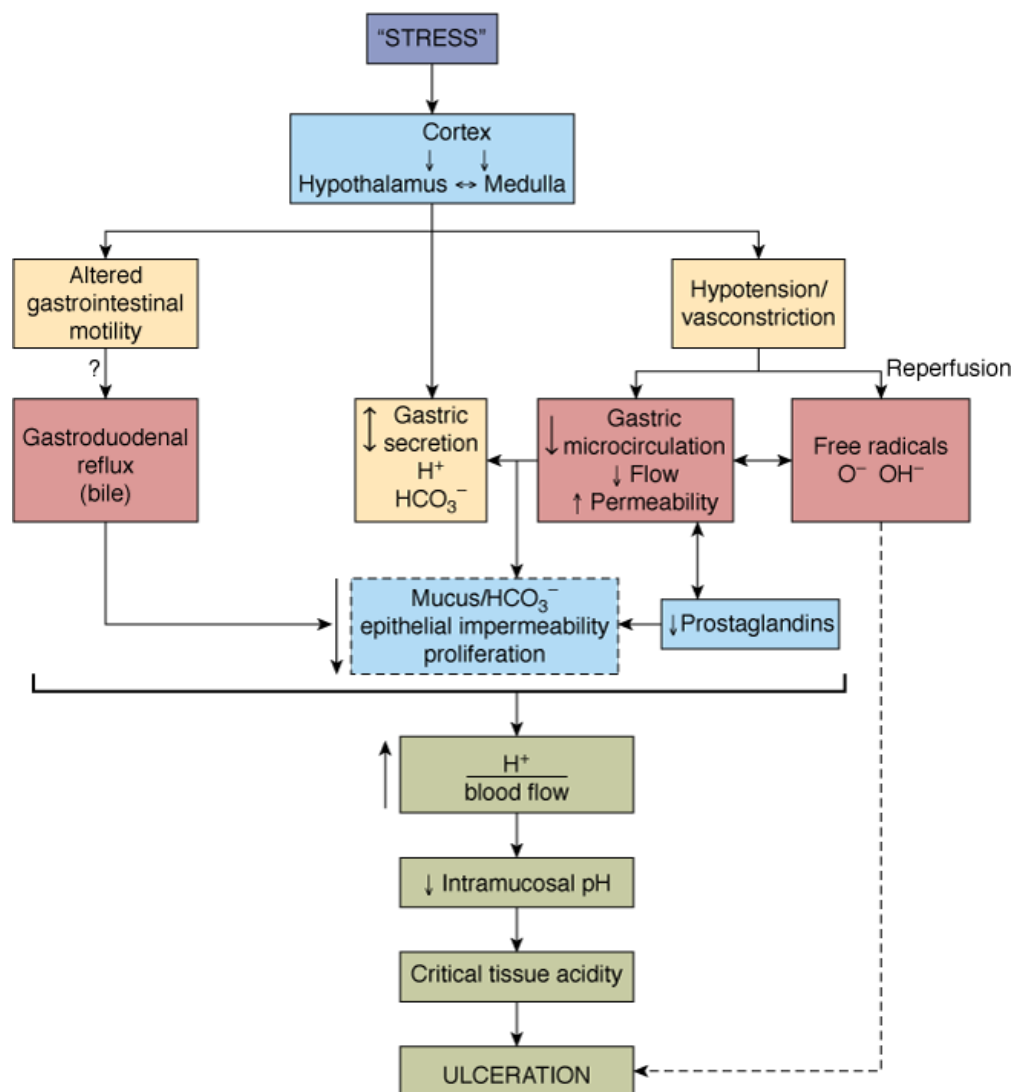
Epidemiology

Stress-related mucosal disease is the most common cause of gastrointestinal bleeding in patients receiving mechanical ventilation. An estimated 74% to 100% of critically ill patients have endoscopically detectable lesions in the gastric mucosa within hours after admission.^{16,17} These lesions are most frequently reported in the acid-producing areas of the stomach in contrast to peptic ulcers, which are more common in the antrum and the duodenal bulbs. Overt stress-related gastrointestinal bleeding occurs in up to 5% of critically ill patients.¹⁸ Other studies have demonstrated an even lower incidence of clinically significant bleeding, ranging from 0.17% to 1.5%.¹⁹ The risk increases with increasing number of days of mechanical ventilation and length of intensive care unit (ICU) stay.²⁰ Other contributing factors include major surgery, head trauma, severe burns, sepsis, glucocorticoids, and renal and hepatic disease on admission.²⁰ Unsurprisingly, clinically significant gastrointestinal bleeding has been linked to prolonged ICU length of stay by as much as 11 days and to a markedly increased mortality, although the primary cause of mortality is attributable to the primary disease process rather than gastrointestinal hemorrhage.²⁰

Pathophysiology

The pathophysiology of stress-related mucosal disease underlines a complex interaction of opposing vectors (Fig. 41-2).²¹ Under normal physiologic conditions, the integrity of the mucosa depends on a delicate balance between injurious factors (gastric acid, enzyme secretion) and protective mechanisms (mucous production, prostaglandins).²² Gastric acid is considered essential for stress ulceration but it is not the sole factor in the pathogenesis of mucosal disease. In the setting of decreased splanchnic blood flow, oxygen radicals are released, prostaglandins synthesis is reduced, and nitric oxide production is exaggerated. These changes perpetuate the release of inflammatory cytokines and triggers cell death.²³ As a result, increased back diffusion of hydrogen ions and pepsin occurs without the mitigating effect of bicarbonate-rich mucous layer and the protective effect of prostaglandins. Collectively, the imbalance between the noxious gastric acid and the impaired reparative mechanisms predisposes patients on mechanical ventilation to stress-related mucosal injury.

Figure 41-2



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Proposed mechanisms for the development of stress ulceration during mechanical ventilation. (Used, with permission, from Bresalier.²¹)

Therapeutic Interventions

The mainstay of the clinical management of stress-related mucosal bleeding in patients who are mechanically ventilated is prevention. Several classes of antistress agents have been introduced since antacids were first evaluated in 1976.²⁴ Antacids directly neutralize luminal gastric acid, bind pepsin and bile acids,²⁵ and may stimulate prostaglandin release from the mucosa.²⁶ The efficacy of antacids in preventing clinically significant stress-related bleeding has been demonstrated in randomized controlled trials.²⁷ The frequent dosing interval needed to achieve and maintain a pH greater than 4, however, is one of the disadvantages of this therapy. In addition to impairing the absorption of other medications, aluminum-containing antacids may be associated with hypophosphatemia and toxic aluminum levels in renal-failure patients, while magnesium-containing antacids may be associated with diarrhea and hypermagnesemia. Sucralfate is another agent that shields the gastric mucosa by forming a protective barrier from the acid in the gastric lumen, stimulating mucous and bicarbonate secretion, and enhancing prostaglandin release.²⁸ A meta-analysis of the efficacy of sucralfate compared with that of histamine-2-receptor antagonists and antacids for the prophylaxis of stress ulcers indicated that sucralfate was at least as effective as the other agents.²⁹ Sucralfate is generally well tolerated but may cause constipation, and aluminium toxicity may occur in patients with renal failure. Misoprostol, a synthetic prostaglandin E_1 analog with gastric cytoprotective and antisecretory properties, has also demonstrated effectiveness in preventing gastric injury and complications induced by nonsteroidal antiinflammatory drugs.³⁰ Its use is limited, however, by the high rate of diarrhea and the need for multiple daily doses.

The introduction of histamine-2-receptor antagonists and proton pump inhibitors eclipsed the use of antacids, and medications in these classes are currently the standard therapy for stress-related mucosal disease. Although continuous infusions may be more effective in suppressing gastric acid,³¹ no studies have demonstrated improved safety, more effective prophylaxis, or a lower rebleeding rate with either method. A major concern of using histamine antagonists is the development of tolerance, which occurs within 42 hours after intravenous administration by both bolus and continuous infusion.³² Histamine antagonists are also associated with adverse chronotropic and inotropic effects and, experimentally, may induce a dose-dependent coronary vasoconstriction.²⁵ The clinical significance of these effects is unclear.

Proton pump inhibitors irreversibly bind the hydrogen-potassium-adenosine triphosphatase, the enzyme responsible for secreting acid into the gastric lumen. Unlike histamine antagonists, proton pump inhibitors do not seem to develop tolerance with sustained therapy. Although both histamine antagonists and proton pump inhibitors can elevate intragastric pH to more than 4, proton pump inhibitors are more likely to maintain a pH above 6.¹⁸ Yet in a meta-analysis of randomized, controlled trials that directly compared proton pump inhibitors with histamine antagonists in prevention of stress-related upper gastrointestinal bleeding in ICU patients, both agents were equally efficacious in preventing stress-related bleeding.³³

The safety consideration of acid-suppressive therapies has been a concern after several studies suggested that acid-suppressive agents may increase the risk of pneumonia.^{34,35} Numerous small studies^{36,37} have examined this issue in ICU patients, but multicenter randomized studies did not demonstrate any significant difference in pneumonia rates between sucralfate and histamine antagonists³⁸ or between proton pump inhibitors and histamine antagonists.³³ Other potential adverse effects for acid-suppressing agents include higher rates of *Clostridium difficile*-associated diarrhea³⁹ and enteric infections.⁴⁰ None of these associations, however, have been confirmed in randomized trials.

Because acid injury may potentiate mucosal ischemic changes, enteral nutrition could potentially decrease stress ulceration by raising intragastric pH. Several studies in mechanically ventilated patients⁴¹ and in burn patients⁴² have associated enteral feeding with a lower incidence of stress-related bleeding. Other studies in critically ill patients, however, have demonstrated that enteral feeding does not have a significant effect on increasing gastric pH and therefore may be ineffective in affording gastroprotection.^{43,44} Definite recommendations regarding the role of enteral nutrition for stress-ulcer prophylaxis are not possible at the present because of lack of prospective, randomized trials.

Motility Disturbances

Several studies indicate that abnormal gastrointestinal motility is common in mechanically ventilated patients.^{45,46} The prevalence of abnormalities in gastric emptying is estimated to be as high as 50%.¹⁵ Using manometric evaluation, it was demonstrated that the contractile activity of the stomach is severely depressed in patients receiving PPV—but persistent, albeit reduced, in the duodenum.⁴⁷ These abnormalities are thought to result from the dysfunction of the interstitial cells of Cajal that act as the controller of gastrointestinal motor activity.⁴⁸ Several factors are implicated in the pathophysiology of altered gut motility, including preexisting diseases, release of endotoxin⁴⁹ and corticotropin-releasing factor⁵⁰ during severe stress, and drugs routinely used in the management of patients on mechanical ventilation (e.g., sedatives and opioid analgesics, catecholamine vasopressors, anticholinergics, and α_2 -adrenergic receptor agonists). Hyperglycemia has been considered as a risk factor to impair pyloric and antral contractions in healthy volunteers.⁵¹ A relationship, however, between hyperglycemia and delayed gastric emptying remains unclear.

Current recommendations for the treatment of impaired gastrointestinal motility focus on optimizing fluid intake, correcting electrolyte disturbances, and minimizing the use of drugs that slow gut motility. Prokinetic agents are not routinely recommended though a recent survey revealed standard prokinetic use in 39% of critically ill patients.⁵² A combination therapy of erythromycin and metoclopramide provides the most effective regimen in improving the delivery of nasogastric nutrition.⁵³ Owing to a rapid tachyphylaxis following erythromycin or metoclopramide administration, however, therapeutic use should be limited to 3 days. Newer therapeutic approaches, such as μ -opioid receptor antagonists and cholecystikinin-1 receptor antagonists, hold promise in alleviating motility disorders but have not been studied so far in critical illness.

If treatment of gastric stasis fails or is contraindicated, the stomach can be bypassed with an intestinal feeding tube. Early enteral feeding can be more successful if feeds are delivered directly to the small intestine. Postpyloric tubes have been shown to be equally effective as prokinetic

treatment in patients who failed nasogastric feeding,⁵⁴ but there is currently no evidence to support routine use of these tubes in critically ill patients.

Acalculous Cholecystitis

Acalculous cholecystitis is a serious and potentially life-threatening illness in critically ill ventilated patients if unrecognized. The condition accounts for 5% to 10% of all cases of acute cholecystitis.^{55,56} The etiology of the disease remains unknown, although PPV for more than 72 hours is considered a risk factor.⁵⁵ Other predisposing conditions include shock, dehydration, multiple transfusions, total parenteral nutrition, and drugs (opiates, sedatives, ceftriaxone) (Table 41-1).^{57,58} Clinical assessment may not be reliable, particularly in intubated and sedated patients. Ultrasonography remains the diagnostic test of choice, although computed tomography has the advantage of being more sensitive in diagnosing acalculous cholecystitis.^{59,60} Evidence of gallbladder wall thickness greater than or equal to 4 mm, pericholecystic fluid or subserosal edema without ascites, intramural gas, or a sloughed mucosal membrane are considered diagnostic criteria for acute acalculous cholecystitis. Hepatobiliary scintigraphy is compromised by frequent false positives and is more helpful in excluding rather than confirming the diagnosis.⁵⁹ The mainstay of therapy for acalculous cholecystitis has been cholecystectomy, but for the critically ill patients percutaneous cholecystostomy is considered an alternative to open procedures.⁶¹

Table 41-1: Predisposing Factors for Acalculous Cholecystitis in Critically Ill Patients

Decreased motility
o Surgery
o Burns
o Mechanical ventilation
o Narcotic analgesics
Decreased cystic artery blood flow
o Arteriosclerosis
o Diabetes
o Shock
o Cardiac failure
Infection
o Salmonella
o Cholera
o Campylobacter
o HIV
Obstruction of cystic duct
o Lymphadenopathy
o Metastatic malignancy

Hepatic Complications

The normal adult liver has a dual blood flow and oxygen supply. Approximately two-thirds of hepatic blood flow and one-half of the oxygen supply is derived from the portal vein while the rest is provided by the hepatic artery. Institution of PPV has significant implications on hepatic perfusion. The reduction in cardiac output observed during PPV causes a proportional drop in global hepatic blood flow.^{62,63} In addition, the descent of the diaphragm during PPV results in a direct compression of the liver parenchyma and a dramatic rise in intraabdominal pressure. The combination of these two forces leads to an increase in hepatic vascular resistance, which, in turn, impedes portal venous flow. Maintenance of spontaneous breathing during airway pressure release ventilation results in higher hepatic venous oxygen saturation and better hepatic lactate elimination as compared with full ventilator support at equal airway pressure limits.⁶⁴

The addition of PEEP further complicates the picture. In clinical studies of patients with acute lung injury secondary to septic shock, an increase in PEEP to 15 cm H2O resulted in decrease hepatic vein oxygen saturation compared to 10 cm H2O.⁶⁵ Similarly, PEEP levels of 15 cm

H₂O but not 10 cm H₂O was associated with a decrease in hepatic glucose production. After liver transplantation, patients are especially vulnerable to change in blood flow because of vascular anastomoses and liver function recovery after cold ischemia. Despite an increase in central venous and pulmonary capillary occluding pressure following 10 mbar of PEEP, there was fortunately no deterioration in Doppler flow velocities of portal and hepatic veins.⁶⁶

Permissive hypercapnia has been observed to increase hepatic and splanchnic blood flow in a biphasic manner. Blood flow is initially reduced because of sympathetic stimulation and is then increased secondary to the direct vasodilator effect of CO₂.⁶⁷ The heterogeneity, however, observed in the individual changes of splanchnic perfusion secondary to decreased V_T supports the concept that the direct local vasodilation of an elevated tissue partial pressure of carbon dioxide (P_{CO₂}) is opposed by the increased release of catecholamines in the systemic circulation, with an end result of no significant change.⁶⁸

Pneumoperitoneum

An association between PPV and pneumoperitoneum has long been described.^{69,70} The mechanism involves airflow dissection through overdistended alveoli into the pulmonic perivascular sheaths.⁷¹ The pocket of air dissects to the mediastinum and migrates through the foramina of Morgagni and Bochdalek to cause free air in the peritoneal space.^{72,73} Risk factors include high airway pressures, large V_T, noncompliant lungs, and preexisting pulmonary disease, including obstructive airway disease and acute respiratory distress syndrome. The diagnosis may be easily mistaken for a perforated viscus. In the absence of leukocytosis, abdominal pain, and peritoneal signs, perforation can be excluded by radiographic imaging with water-soluble contrast material administered via oral, rectal, or enterostomy tube.^{74,75} Other approaches to define the etiology of pneumoperitoneum include aspiration and analysis of the partial pressure of oxygen (P_{O₂}) of the intraperitoneal free gas.^{76,77} Spontaneous resolution usually occurs within a few days.^{62,78} On rare occasions, “tension pneumoperitoneum” has been described, in which surgical decompression is necessary, even in the absence of peritoneal signs, to relieve vascular collapse.⁷⁹

Cardiovascular Complications

Cardiovascular complications are associated with critical illness and mechanical ventilation, but are rarely a direct complication. The most common complications include arrhythmias, myocardial ischemia, usually a type II or demand non-ST-segment elevation myocardial infarction, although primary acute coronary syndrome (ST-segment elevation myocardial infarction) may also occur, as most patients have antecedent risk factors for coronary artery disease.

Arrhythmias

Most arrhythmias in medical ICU patients are tachyarrhythmia (90%), which are divided almost equally between supraventricular (atrial fibrillation 30%) and ventricular foci (monomorphic V_T 49%).⁸⁰ In surgical ICU patients, 61% of tachyarrhythmias were atrial fibrillation.⁸¹ Although arrhythmias may occur as a result of underlying structural heart disease, physiologic alterations associated with the presenting illness and intrathoracic changes associated with PPV may also contribute. Ventilated patients often experience significant hypoxemia and hypercapnia, acidemia or alkalemia, hypokalemia, hypomagnesemia, and hypocalcemia that can all precipitate arrhythmias. Arrhythmias can also be precipitated by many drugs: vasopressors, inotropes, and inhaled β-agonists being the most common offenders. Clinicians must weigh the risk-to-benefit ratio of the drug relative to the degree of arrhythmia: specifically, is the arrhythmia more likely to have been triggered by the bronchospasm, hypoxemia or hypercapnia, or a β-agonist? Impaired electrical conduction of the normal cardiac impulse has obvious systemic effects, leading to decreased cardiac output and compounding the preexisting respiratory compromise. Identification and management of the inciting cause of the arrhythmia is important, as is appropriate management, as directed by advanced cardiac life support guidelines of the American Heart Association.⁸²

Myocardial Ischemia

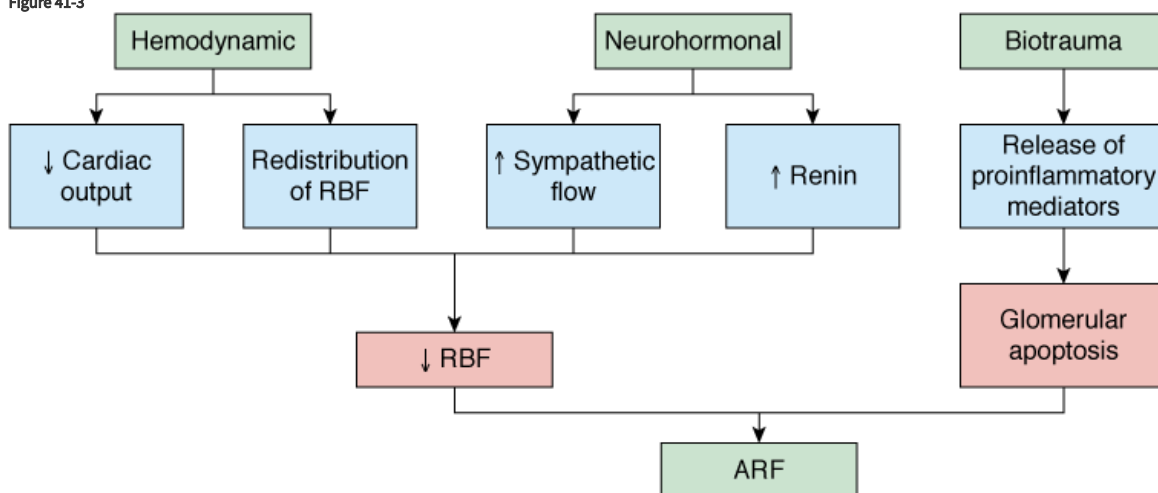
Myocardial ischemia in ventilated patients can be primary acute coronary syndrome (plaque rupture) or demand-related ischemia (non-ST-segment elevation myocardial infarction) in patients requiring mechanical ventilation for noncardiac indications. In patients presenting with acute coronary syndrome, the need for mechanical ventilation has been associated with increased mortality rates (as high as 50%).⁸³ Myocardial ischemia can occur at any time during the period of mechanical ventilation, and the increased systemic and myocardial oxygen

demand during the time of weaning trials may precipitate acute ischemia.^{84–88} Diagnosis is complicated by the lack of classical symptoms of chest pain and the low sensitivity of continuous electrocardiogram (ECG) monitoring.⁸⁹ The diagnosis should be considered in patients failing weaning trials. Diagnostic criteria should include clinical, ECG, and cardiac biomarkers, recognizing the limitations associated with biomarker use in the ICU (troponin elevation in clinical presentations of right-heart strain and elevated pulmonary vascular resistance, false elevations in renal failure). Treatment is based on advanced cardiac life support guidelines,⁹⁰ including interventional procedures.

Renal Complications

The first report to show the impact of PPV on renal function was published in 1947.⁹¹ Since then, several studies have documented an association between mechanical ventilation and development of renal failure in the ICU setting.^{92–94} PPV increases the odds of developing renal failure threefold when PEEP is below 6 cm H₂O and more than 17-fold when PEEP is above 6 cm H₂O, despite volume replacement, maintenance of normal filling pressures, and adequate oxygen delivery.⁹² Various mechanisms have been proposed to explain the alterations in renal function in patients receiving PPV, although three mechanisms are judged dominant: hemodynamic, neurohormonal, and biotrauma (Fig. 41-3).

Figure 41-3



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Mechanisms of acute renal failure during mechanical ventilation. ARF, acute renal failure; RBF, renal blood flow.

The systemic hemodynamic effects of PPV originate in a complex interaction between intrathoracic pressure, intravascular volume, and cardiac output. The increase in intrathoracic pressure has been shown to correlate with a decrease in renal plasma flow, glomerular filtration rate, and urine output.^{95,96} The reported effects, however, of PPV on glomerular filtration rate and renal blood flow are variable and may reflect differences in hydration status, underlying pulmonary and cardiac dysfunction, and use of vasoactive agents.⁹⁷ It has been suggested that redistribution of intrarenal blood flow from the cortical to juxtamedullary nephrons may play an important role in causing reduction of renal function during PEEP.⁹⁸ This may lead to decreased urinary output and creatinine clearance suggesting that small decreases in renal blood flow can seriously affect renal function.

PPV induces several neurohormonal changes in the sympathetic outflow, the renin–angiotensin axis, nonosmotic vasopressin release, and the atrial natriuretic peptide production. Thus far, no definite correlation between antidiuretic hormone levels^{99,100} or atrial natriuretic factor,^{101,102} and renal function during PPV has been confirmed. PPV, however, increases plasma renin levels by twofold,^{96,102} driven by an increase in sympathetic tone. Consequently, renal blood flow and glomerular filtration are reduced, causing^{103,104} a drop in distal tubule delivery of sodium. This, in turn, leads to further activation of the renin–angiotensin–aldosterone axis and increased sodium avidity. Clinically, these changes are manifested by a decrease in urine output as a result of decreased osmolar excretion. There are no definitive therapeutic trials for the treatment of PPV-induced renal hypoperfusion. Several small trials have shown that fluid administration and the use of vasoactive drugs (dopamine at 5 µg/kg/min or fenoldopam) may improve renal function,^{101,105} but results are not conclusive. It is unlikely that approaches focused solely on the hemodynamic and neurohormonal mechanisms of PPV-induced renal dysfunction will be clinically effective in preventing or treating this multifactorial problem.

In addition to altering renal blood flow, PPV can impact renal function through the release of proinflammatory cytokines. These include IL-8, IL-6, and TNF- α , which promote glomerular and tubulointerstitial sequestration of neutrophils, upregulation of leukocyte adhesion molecules, and a decrease in filtration fraction associated with alterations in vascular tone.^{106–111} Moreover, injurious ventilation strategies, such as high V_T and low PEEP, induces glomerular cell apoptosis via a soluble FasL-mediated pathway.¹⁰⁷ Regardless of the proposed mechanism, it is clear that derangements in the innate immune/inflammatory response, oxidative stress and cellular necrosis/apoptosis are important components of this organ cross talk in response to injury.

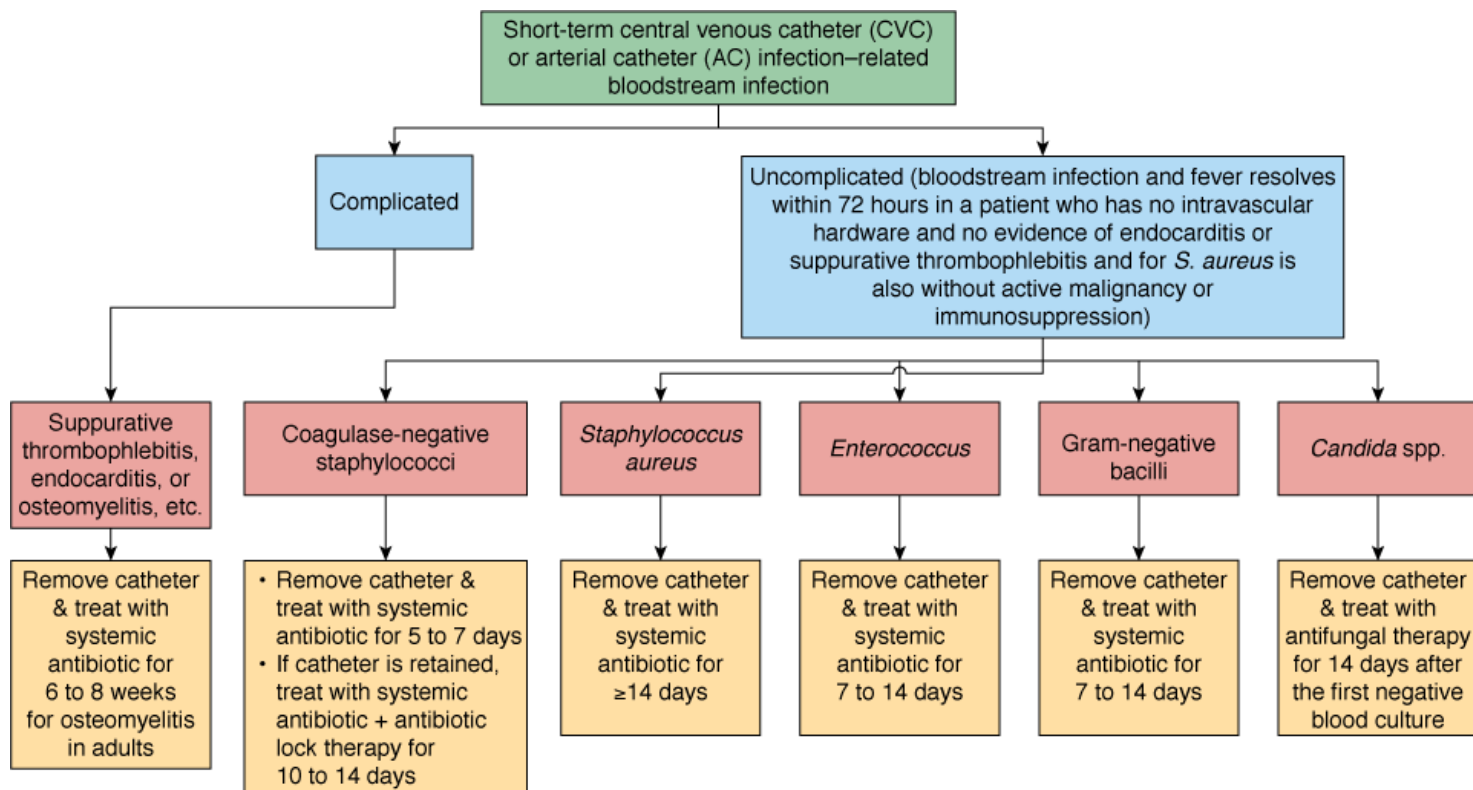
Several authors have described the effects of other ventilation techniques on renal function. Comparison of controlled mechanical ventilation plus PEEP with low-frequency ventilation and extracorporeal carbon dioxide removal resulted in an immediate increase in urinary flow, osmolar clearance, and creatinine clearance.¹¹² Spontaneous breathing during airway pressure release ventilation provides improved systemic blood flow and regional perfusion to the kidneys.¹¹³ The effect of permissive hypercapnia on renal function is also well documented: Partial pressure of arterial carbon dioxide (P_{aCO_2}) levels correlate inversely with renal blood flow.¹¹⁴ Hypercapnia causes renal vasoconstriction directly¹¹⁵ and stimulates noradrenaline release through activation of the sympathetic nervous system.¹¹⁶ Indirectly, hypercapnia induces systemic vasodilation, which results in a drop of the systemic vascular resistance and subsequent activation of the renin–angiotensin–aldosterone system.¹¹⁷ As a result, renal blood flow becomes compromised and worsen glomerular filtration rate. These hypocapnic changes occur independently of partial pressure of arterial oxygen (P_{aO_2}),¹¹⁸ suggesting that P_{aCO_2} plays a pivotal role in determining the vascular response to changes in blood-gas pressures.

Infectious Complications

Most clinical investigations indicate that 25% to 50% of ICU patients experience one or more nosocomial infection.^{119–121} Fever is the usual trigger for a set of diagnostic and therapeutic interventions. Although the source of fever is not always identified, ventilator-associated infection, sinusitis, catheter-related bacteremias, nosocomial diarrhea, and wound infections account for most infections.¹²² Two of these complications (i.e., ventilator-associated pneumonia and sinusitis) are discussed in [Chapters 46](#) and [47](#).

Infection of central venous catheters is a particular problem with a mortality of 10% to 35%.¹²³ The absolute number of catheter-related infections is rising annually because of increasing use, although incidence per catheter days has actually decreased.¹²⁴ The catheter insertion site itself provides the most direct route of entry for the pathogen and this is the most common cause of infection.¹²³ These infections are caused mainly by gram-positive bacteria, in particular *Staphylococcus aureus* and coagulase-negative staphylococci such as *Staphylococcus epidermidis*.¹²⁵ Infections, however, can be caused by a wide range of microorganisms including *Enterococci*, *Candida* spp., *Acinetobacter* spp., *Pseudomonas* spp., and *Klebsiella* spp.¹²⁶ Catheter management depends on the likely specific pathogens and the individual colonization profile of the patient ([Fig. 41-4](#)).¹²⁶

Figure 41-4



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Approach to the management of patients with central venous catheter-related or arterial catheter-related bloodstream infection. (Reprinted with permission from Mermel LA, Allon M, Bouza E, et al. Clinical practice guidelines for the diagnosis and management of intravascular catheter-related infection: 2009 Update by the Infectious Diseases Society of America. *Clin Infect Dis*. 2009;49:1–45 with permission of Oxford University Press.)

Infectious causes of nosocomial diarrhea include *Pseudomonas aeruginosa*, *Escherichia coli*, and cryptosporidium, although *C. difficile* accounts for the majority of hospital-acquired diarrhea or colitis.¹²⁷ Prior antibiotics, advanced age, gastric acid suppression, and gastrointestinal surgery or manipulation are major risk factors for symptomatic infection. At least 20% of adults ventilated for more than 1 week are colonized with *C. difficile* and up to 4% develop infection.¹²⁸ A hypervirulent strain, the North American pulse-field gel electrophoresis type 1 (NAP1) strain, has been implicated as a cause of *C. difficile* infection of increasing severity.^{129,130} A deletion in the toxin regulatory gene, *tcdC*, is thought to allow this strain to overproduce toxin A and toxin B by as much as 15- to 20-fold.¹²⁹ Outbreaks caused by this strain are strongly associated with use of fluoroquinolones, although other antibiotics are also implicated.¹³¹ For *C. difficile* infection of every severity, cessation of the inciting antibiotic is the first step in management whenever possible. Metronidazole and vancomycin are the antibiotics most commonly used to treat *C. difficile* in patients with symptomatic infection. A novel macrocycle antibiotic, OPT-80, is currently in phase 3 trials. OPT-80 is minimally absorbed from the gastrointestinal tract and well tolerated in most subjects.¹³² Although highly effective against *C. difficile*, OPT-80 leaves the majority of the gram-negative anaerobic flora of the gastrointestinal tract intact. Another commonly employed strategy is the administration of probiotics. In several randomized controlled trials,^{133–135} however, the combination of *Saccharomyces* or *Lactobacillus* probiotics with conventional antibiotic therapy has failed to show significant benefit.

Hematologic Complications

Anemia

Ninety-five percent of ventilated patients develop anemia after the third ICU day.¹³⁶ A significant part is attributed to frequent and excessive blood draws. ICU patients are phlebotomized 3.4 times a day (mean volume of 41.5 mL of blood) as compared to 1.1 times a day for patients on a general ward.¹³⁷ Occult gastrointestinal bleeding represents another source of blood loss. The condition is aggravated by coagulation disorders that include platelet dysfunction, thrombocytopenia, loss of coagulation factors, and endothelium-related coagulation disorders.¹³⁸

Several investigations in the last decade, however, suggest that anemia in ICU patients bears many similarities to that associated with chronic disease. More than 90% of ICU patients have low serum iron, total iron-binding capacity, and iron-to-total-iron-binding-capacity ratio, but have a normal or, more usually an elevated, serum ferritin level.^{139,140} At a time when the iron studies are abnormal, serum erythropoietin levels are only mildly elevated with little evidence of reticulocyte response to endogenous erythropoietin.¹³⁹ Various proinflammatory cytokines are implicated in this blunted response. Inhibition of renal erythropoietin-gene expression in rat kidneys has been demonstrated for IL-1 and TNF- α .¹⁴¹ In some cases, a shorter erythrocyte life span may aggravate the anemia. There are strong indications that the mononuclear phagocytic system plays a key role, with enhanced iron entrapment and a temporary inhibition of iron release,¹⁴² while the bone marrow is still capable of incorporating iron and of reacting to exogenous erythropoietin.

Critically ill patients respond to treatment with recombinant human erythropoietin and parenteral iron with increased erythropoiesis.¹⁴³ Despite early promising results,^{144,145} two large placebo-controlled trials investigating the exogenous erythropoietin in critically ill patients failed to show improved 28-day mortality, shorter duration of mechanical ventilation, decreased ICU length of stay, or reduction in red-cell transfusions.^{143,146} There was, however, an increased propensity for thrombotic vascular events in the erythropoietin arm.¹⁴⁶

Thrombocytopenia

The incidence of thrombocytopenia in ICU patients varies from 23% to 41%, depending on the definition of thrombocytopenia and the ICU type.¹⁴⁷⁻¹⁴⁹ Thrombocytopenia is often multifactorial (Table 41-2) and a marker of disease severity.¹⁵⁰ Nearly all studies have found an inverse correlation between platelet count and prolonged ICU stay or hospital mortality.¹⁵¹⁻¹⁵³ The magnitude of the drop in platelet count correlates more closely with adverse outcomes than the nadir in absolute count. The platelet count pattern over time provides important clues as to the cause of the thrombocytopenia. A gradual decrease in platelet counts over several days suggests worsening of the underlying disease and is often associated with multiorgan failure, whereas a new, rapid decrease in platelet count that begins after the fourth day is typical for immune-mediated causes. A classic example is heparin-induced thrombocytopenia, which is frequently considered as a cause of thrombocytopenia in ICU patients, although its incidence is only 0.3% to 0.5%.^{154,155} It occurs in two forms. With the more common, type I, form, which occurs in 10% to 20% of patients receiving unfractionated heparin, nonimmune mechanisms produce a drop in platelet counts 1 to 4 days after the onset of therapy. The type I form is not associated with hemorrhagic or thrombotic sequelae, and management involves observation; platelet counts normalize in most patients despite heparin continuation.¹⁵⁶ With the type II form, in contrast, 30% to 80% of patients experience thrombotic sequelae. Venous thromboses outnumber arterial thromboses by 4:1, and life-threatening pulmonary embolism occurs in 25% of patients who have thrombosis.¹⁵⁶ The type II form occurs in 1% to 3% of patients who receive unfractionated heparin, and is induced by antibodies of the immunoglobulin G class against platelet factor 4-heparin complexes.¹⁵⁶ Heparin-induced thrombocytopenia is suspected when a platelet count falls by more than 50% after 4 days of treatment, and can present with new thrombosis, typically occurring 5 to 14 days after the start of prophylactic or therapeutic doses of heparin. To prevent new thrombosis, nonheparin anticoagulant therapy is required. Four drugs are approved for this purpose: (a) two direct thrombin inhibitors, lepirudin and argatroban; (b) a heparinoid, danaparoid; and (c) an antifactor Xa inhibitor, fondaparinux.

Table 41-2: Etiology of Thrombocytopenia in Mechanically Ventilated Patients

Hemodilution	• Large volume of fluid infusion - Excessive transfusion of packed red cells or plasma exchange
Increased platelet consumption	• Sepsis • Thrombotic thrombocytopenic purpura • Disseminated intravascular coagulopathy • Hyperfibrinolysis (metastatic prostate/ovarian cancer) • Intravascular devices
Decreased platelet production	• Myelodysplasia or leukemia • Drug-induced bone marrow depression • Radiation
Increased platelet destruction	• Autoimmune thrombocytopenia • Drug-induced thrombocytopenia • Posttransfusion purpura
Sequestration	• Hypersplenism • Hypothermia

Neuromuscular Complications

Critical illness polyneuropathy is an acute axonal sensory motor polyneuropathy, mainly affecting the lower limb nerves of critically ill patients including those receiving mechanical ventilation. Critical illness polyneuropathy is often associated with critical illness myopathy and differentiating between these entities may be difficult. For both, the clinical presentation is one of limb weakness or flaccidity, and respiratory muscle weakness or persistent respiratory failure, manifesting as unexplained difficulty in ventilator weaning. Most patients have an antecedent diagnosis of sepsis or septic shock, and the diagnosis should be considered in all patients with difficult weaning. Occurrence varies from 58% of patients with prolonged ICU stay,^{157,158} to 70% to 80% of patients with sepsis, septic shock, or multisystem organ failure,^{158–162} to 100% of patients with sepsis and coma.^{158,163}

Etiology

The cause is believed to be microvascular damage, depletion of bioenergetic neuron reserves, altered sodium channel inactivation, and altered glycemic control, mediated by inflammatory cytokines,^{158,162,164–170} and an association with use of glucocorticoids.¹⁷¹ Although previous data ascribed increased risks with the use of depolarizing neuromuscular blockade and aminoglycoside administration, more recent data suggests that their association may be related to underlying severity of illness.^{171–173} A multivariate analysis has identified four independent risk factors: female gender (odds ratio [OR] 4.66), number of days with dysfunction in equal to or greater than two organs (OR 1.28), duration of mechanical ventilation (OR 1.10), and corticosteroid administration (OR 14.9).¹⁷¹

Diagnosis

Electrophysiology is required to make the definitive diagnosis in distinguishing between polyneuropathy, myopathy, and neuromuscular blockade, as clinical neurologic examination is insensitive. These studies, however, can be difficult to obtain. Evaluation usually includes needle electromyography in upper and lower limbs. Phrenic nerve conduction studies and needle electromyography of the respiratory muscles may suggest critical illness polyneuropathy as a cause of difficult weaning. The classic electromyography pattern in critical illness polyneuropathy is one of primary axonal degeneration, and manifests as a reduction in the amplitude of the compound action potentials and

sensor nerve action potentials. A decrease in the amplitude and prolonged duration of compound action potentials, which suggests an associated myopathy, may be one of the first signs of a developing polyneuropathy.^{174,175} Additional features include inexcitability of the muscle on direct stimulation and abnormal spontaneous electromyographic activity. The decline in sensory nerve conduction may present after the above findings. Electrophysiologic studies may help to differentiate between generalized neuromuscular disorders associated with critical illness (Table 41-3); however, these studies require fully cooperative patients. Those afflicted do not necessarily have a fatal outcome as had previously been ascribed^{163,176} to comatose patients with acute paralysis, because many patients make a full or near-full recovery.^{158,162}

Table 41-3: Generalized Neuromuscular Conditions Encountered in Mechanically Ventilated Patients

Condition	Clinical Findings	Electrophysiologic Findings	Creatine Phosphokinase	Muscle Biopsy	Prognosis
Polyneuropathy					
Critical illness polyneuropathy	Flaccid limbs, respiratory weakness	Axonal degeneration of motor and sensory fibers	Nearly normal	Denervation atrophy	variable
Neuromuscular Transmission Defect					
Transient neuromuscular blockade	Flaccid limbs, respiratory weakness	Abnormal repetitive nerve stimulation studies	Normal	Normal	good
Critical Illness Myopathy					
Thick filament myosin loss	Flaccid limbs, respiratory weakness	Abnormal spontaneous activity	Mildly elevated	Loss of thick (myosin) filaments	good
Rhabdomyolysis	Flaccid limbs	Nearly normal	Markedly elevated	Marked necrosis	good
Necrotizing myopathy of intensive care	Flaccid weakness, myoglobinuria	Severe myopathy	Markedly elevated (myoglobinuria)	Marked necrosis	poor
Disuse (cachectic) myopathy	Muscle wasting	Normal	Normal	Normal or type II fiber atrophy	good
Combined polyneuropathy and myopathy	Flaccid limbs, respiratory weakness	Indicate combined polyneuropathy and myopathy	Variable	Denervation atrophy and myopathy	variable

Source: Adapted, with permission, from Bolton.¹⁷⁵

Prognosis

Patients with severe respiratory and limb weakness with electromyogram evidence of predominant muscle involvement, minimal elevations of creatine phosphokinase and normal muscle biopsy may make a rapid recovery, whereas patients with significant enzyme elevation and evidence of muscle necrosis may have lasting deficit.¹⁷⁵

Nutritional Support

Appropriate nutritional support in ventilated patients has emerged as an important variable in outcomes of ICU patients. The metabolic response to stress and injury is characterized by increased release of cytokines (IL-1, TNF- α , IL-6) with increased production of counterregulatory hormones (catecholamines, cortisol, glucagon, and growth hormones). Counterregulatory hormones induce catabolism and oppose the anabolic effects of [insulin](#). The resultant effects are hypermetabolism and hypercatabolism with a loss of body energy stores through proteolysis, lipolysis, and glycogenolysis.^{177–179}

Adverse Effects of Malnutrition

The adverse effects of malnutrition on ventilated patients include altered ventilatory drive; reduction in the ventilatory response to hypoxia; decreased mass; force contractility and endurance of the diaphragm; decreased respiratory muscle strength; hypercapnia; reduced synthesis of alveolar surfactant; and altered humoral and cellular immunity.¹⁸⁰ Despite the general consensus that nutritional support is important, and guidelines advocating provision of enteral nutrition within 24 to 48 hours of ICU admission,^{181–184} observational studies reveal that up to 40% of critically ill patients receive no nutritional support during their ICU stay and 60% of patients in the ICU for at least 3 days remain without nutritional support for 48 hours or longer.^{183,185–187} A recent meta-analysis of enteral nutrition in critically ill patients¹⁸⁶ demonstrated a reduction in mortality and pneumonia attributable to the inception of standard enteral nutrition within 24 hours of injury (trauma, burn) or ICU admission. Although five of the six trials were nonmedical critically ill patients,^{188–192} the one medical trial focused on ventilated patients.¹⁹³ Earlier trials have demonstrated that failure to deliver adequate nutritional support or delaying nutritional support leads to cumulative energy deficit, which correlates with longer ICU stay, increased duration of mechanical ventilation, and more infectious complications.^{194,195}

Recognition of the effects of undernutrition has led to the recent development of screening guidelines, as well as disease-specific nutritional guidelines.¹⁹⁶ Nutritional support studies have been challenged by major methodological problems in terms of defining the populations studied, standardization of severity of illness scoring, metabolic derangements, variability in enteral nutritional formulations, and outcome measures. The European Society for Clinical Nutrition and Metabolism generated the *ESPEN Guidelines on Enteral Nutrition* in 2006,^{181,197–202} and the *ESPEN Guidelines on Parenteral Nutrition* in 2009;^{184,203–206} the American Society for Parenteral and Enteral Nutrition generated the *ASPEN Clinical Guidelines* in 2009.²⁰⁷ The guidelines are similar in their recommendations ([Table 41-4](#)), with some notable differences. The ASPEN guidelines advocate withholding enteral nutrition in patients with hemodynamic compromise or high-dose catecholamine requirements. This data is derived from mixed human and animal models and although splanchnic perfusion is increased with enteral nutrition,^{208–216} multiple animal studies suggest differential splanchnic vasoconstriction with vasopressor agents (vasopressin > dopamine > [phenylephrine](#)). Of note, norepinephrine, the leading vasopressor for septic shock, and dobutamine are both associated with improved gut mucosal pH or perfusion.^{217–219} Consequently, the decision to initiate enteral feedings in cases of hypotension is left to clinical judgment to weigh the risks of bowel ischemia against the benefits of enteral nutrition.

Table 41-4: Guidelines for Enteral Nutrition in Critically Ill Patients

Population	Enteral Nutrition Summary Recommendations
Critically Ill (Nonsurgical)	1. All patients not expected to be on oral diet within 3 days should receive enteral nutrition (EN) 2. EN should be started within 24 hours of admission to ICU 3. 20 to 25 kcal/kg body weight (BW)/day replacement during the acute phase of illness 4. 25 to 30 kcal/kg BW/day replacement during anabolic recovery phase 5. No significant data to support immune-modulating EN formulas or supplements with the exception of: * Acute respiratory distress syndrome (ARDS)—consider EN enriched with omega-3 fatty acids 6. No significant difference in efficacy of jejunal as compared to gastric feeding 7. Parenteral nutrition should be avoided in patients who can tolerate EN and achieve target intake
Gastroenterology	
Liver Disease Alcoholic Steatohepatitis (excludes NASH)	1. Whole protein formulations are generally recommended. Oral nutritional supplements (ONS) are recommended to supplement oral intake 2. Tube feeding (TF) are recommended if oral intake remains inadequate (even if esophageal varices are present) 3. Use branched-chain amino acid EN formulations in patients in whom hepatic encephalopathy begins during EN 4. Consider the use of concentrated high-energy EN formulations in patients with ascites to avoid positive fluid balance 5. Percutaneous endoscopic gastrotomy (PEG) placement is associated with higher complication rates because of ascites and varices 6. There is no increased risk of bleeding with fine bore nasogastric tubes 7. Recommended energy intake: 35 to 40 kcal/kg BW/day 8. Recommended protein intake: 1.2 to 1.5 g/kg BW/day
Cirrhosis	All of the above and: 1. The use of branched-chain amino acid supplements may improve clinical outcome in advanced cirrhosis
Fulminant Liver Failure	1. Patients should receive enteral nutrition via nasoduodenal tube, with close observation for hypoglycemia 2. No recommendations for specific EN formulations (absence of data) 3. Calorie recommendations as per critical illness (absence of data) 4. Glucose, lactate, triglyceride, and ammonia levels should be monitored as used as surrogate markers of substrate utilization
Renal Failure	

Population	Enteral Nutrition Summary Recommendations
Acute renal failure	1. Standard critical care EN formulations are adequate, but patients should be assessed individually 2. If there are significant electrolyte abnormalities, EN formulations for chronic renal failure may be advantageous 3. Nonprotein nutritional requirements: 20 to 30 kcal/kg BW/day as carbohydrate 3 to 5 g/kg BW/day, fat 0.8 to 1.2 g/kg BW/day 4. Protein requirements differ depending on clinical status: a. conservative therapy 0.6 to 0.8 g/kg BW/day b. extracorporeal therapy 1.0 to 1.5 g/kg BW/day c. continuous renal replacement therapy (CCRT), hypercatabolism: up to maximum of 1.7 g/kg BW/day 5. Parenteral micronutrient supplementation in prolonged CRRT can be considered, with vitamin C 30 to 100 mg/day, folate, magnesium, calcium, selenium, and thiamine

Complications of Nutritional Support

Complications of enteral nutrition in ventilated patients include improper tube placement, aspiration (more related to supine positioning because gastric residual volumes have not consistently correlated with aspiration risks^{220,221}), vomiting, diarrhea, abdominal distension, and constipation. Most of these problems can be avoided or managed by altering the enteral formulation and with the use of promotility and stool-bulking agents.^{53,222–231} The presence of gastric residual volumes of 250 to 500 mL in two consecutive measurements over 8 hours should prompt the use of a prokinetic agent; if gastric residual volume consistently exceeds 500 mL, interruption of enteral nutrition should be stopped and jejunal feedings considered.²³²

Parenteral Nutrition

Parenteral nutrition should be reserved for patients unable to tolerate or attain their targeted energy requirements within 2 days of enteral feeding, and patients not expected to be on normal nutrition within 3 days.¹⁸⁴ In some meta-analyses,^{233,234} parenteral nutrition has been associated with increased rates of infectious complications, which may be associated with the hyperglycemia that is seen more often with parenteral than with enteral nutrition. Parenteral nutrition is administered through a dedicated central venous device, given the osmolality. Low-osmolality (<850 mOsm/L) mixtures, designed to cover only a portion of the nutritional needs (usually supplementing enteral nutrition), can be infused through peripheral venous access devices.¹⁸⁴ Lipid emulsion is required as a source of energy as well as essential fatty acids (linoleic [omega-6] and α -linolenic [omega-3] fatty acids). The meta-analysis that suggested an increased complication rate with the use of lipid emulsion was thought to have suboptimal control for total calories and carbohydrates, as per the ESPEN committee on parenteral nutrition, which may have negatively affected outcome. The evidence for a detrimental effect of lipids was not strong.¹⁸⁴ More recent data suggest that delaying the use of lipids does not change complication rates.²³⁵ The choice of long-chain triglycerides from soybean oil or mixtures of long-chain and medium-chain triglycerides from coconut oil is at a practitioner's discretion. Evidence supports tolerance of the mixed emulsions and several small studies have demonstrated clinical advantages (less immunosuppression and fewer infections,^{236–238} and lower rates of oxygen consumption in ventilated patients) than with long-chain triglycerides alone, although prospective controlled studies are lacking. Olive oil-based formulations are also available and thought to be safe and well tolerated in other patient populations (burn, home total-parenteral nutrition for intestinal failure), but only one small retrospective observational trial has been done in critically ill patients. There is conflicting data on the addition of eicosapentaenoic acid and docosahexaenoic acid infusions to lipid emulsion, but may decrease length of stay in critically ill patients.

Venous Thromboembolism

Incidence and Risk Factors

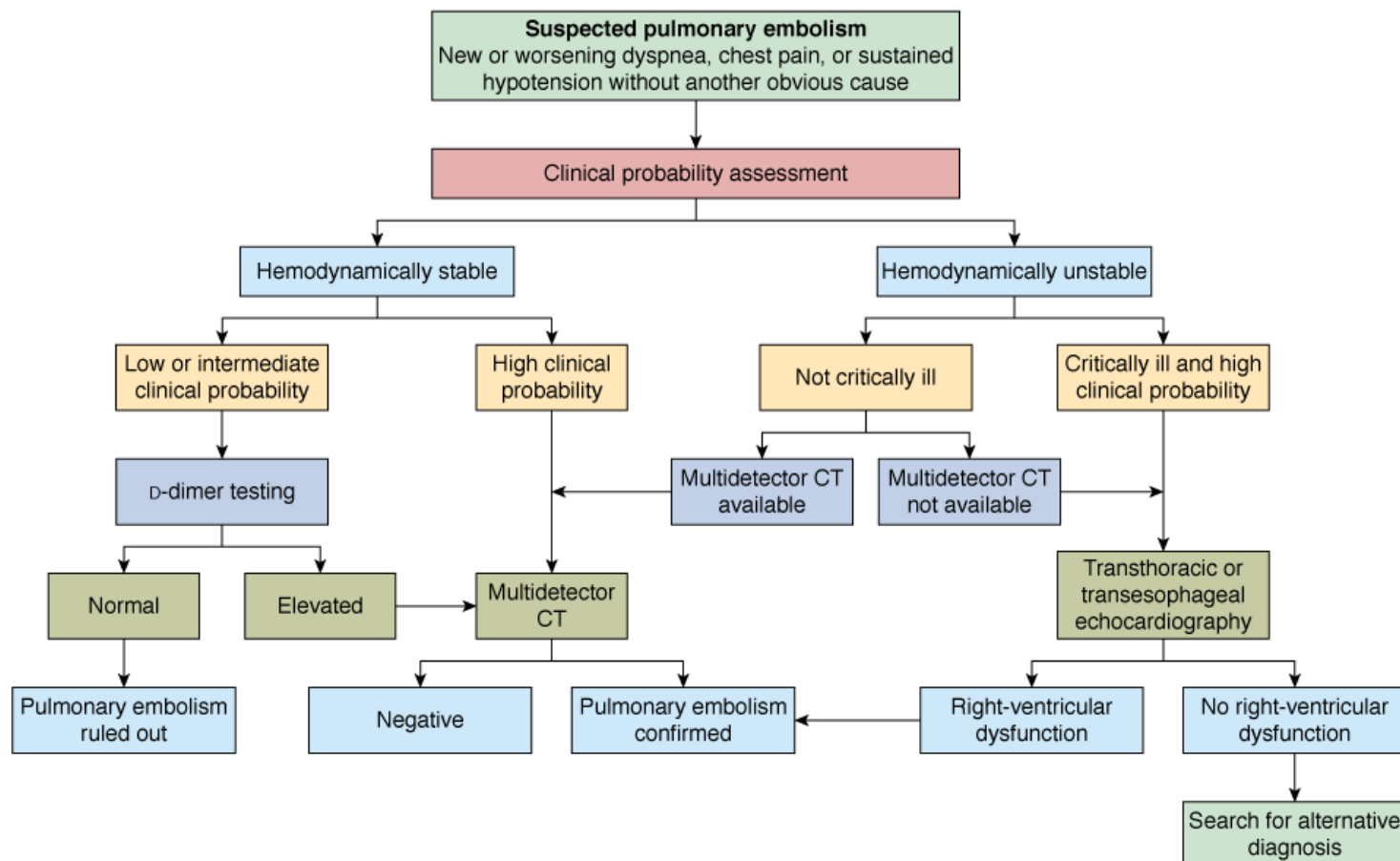
The clinical manifestations of venous thromboembolism (VTE) are protean, ranging from asymptomatic tachycardia and mild dyspnea to hemodynamic collapse. Given this range of presentation, the wide case fatality rate (1% to 60%) is not surprising.²³⁹ Deep venous thrombosis (DVT) is the leading risk factor for the development of VTE, and proximal DVT is found in up to 60% of critically ill patients with venous thrombosis; more proximal thromboses are associated with a 40% to 95% incidence of pulmonary embolism (PE).^{240–243} DVT can be clinically silent, and 10% to 100% of ultrasonographically detected thromboses were not detected on clinical examination.^{241,244–246} There are differences in incidence of DVT in different ICU settings: medical and surgical ICU patients not receiving thromboprophylaxis have an incidence of approximately 30% within 2 weeks, trauma patients 60%, orthopedic surgery patients 40% to 60%, general neurosurgical patients 20% to 50%, and spinal cord injury patients 50% to 80%.^{240,247,248} Risk factors for the development of venous thrombosis include malignancy, duration of mechanical ventilation, immobility (associated with use of sedatives and paralytic agents), severity of illness, emergent surgery, vascular injury from central venous catheterization (femoral), prior VTE, female gender, obesity, cardiac failure, acute myocardial infarction, and inadequate implementation of thromboprophylaxis.^{244,249–251}

Diagnosis

PE should be suspected in all ventilated patients with new or unexplained hypoxemia, tachypnea, or sustained hypotension without an obvious alternative diagnosis. It should be considered in the evaluation of unexplained fever and progressive right heart failure.

The diagnostic approach varies with the severity of illness and is tailored to a patient's hemodynamic stability. The ultimate goal is to objectively define the presence or absence of VTE (Fig. 41-5). In ventilated patients who are hemodynamically stable, the diagnostic approach includes pretest probability evaluation and multidetector CT scanning. d-dimer is excluded from this evaluation. Unlike in the evaluation of outpatients, d-dimer is of little value in hospitalized patients, as it lacks specificity in oncology patients, hospitalized patients, pregnant women, the elderly, and in situations where the pretest probability of VTE is high.²⁵² Most probability scoring systems (Wells Score, Modified Wells Score, Geneva Score, revised Geneva Score, Christopher Study, PERC [Pulmonary Embolism Rule-out Criteria], and PISA-PED [Prospective Investigative Study of Acute Pulmonary Embolism Diagnosis]) are designed for outpatient use or have been evaluated in only small populations of noncritically ill inpatients.^{253–257} If the pretest probability of PE is anything but low, a patient should proceed to CT-angiography, which has been well validated with respect to pulmonary angiography^{258–260} and has an accepted negative predictive value of 95%. Ventilation-perfusion scanning has limited utility in ventilated patients.

Figure 41-5



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
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Diagnostic work-up for suspected pulmonary embolism. (Used, with permission, from Agnelli.²⁶⁰)

The diagnostic approach for hemodynamically unstable patients is not as clearly defined. The focus here is to rapidly exclude embolism-associated obstructive shock from the differential diagnosis. A recently published algorithm divides the hemodynamically unstable group into critically ill and not critically ill, but this can be a difficult distinction.²⁶¹ If the patients can be transported for CT-angiography, they should undergo this procedure, which has a 97% sensitivity for detecting emboli in the main pulmonary arteries.²⁵³ If they are too unstable for transport and they have a high pretest probability of PE, they can undergo either transthoracic echocardiography or transesophageal echocardiography, with the goal of detecting right-ventricular dysfunction (defined as a ratio of right-ventricular-to-left-ventricular end-diastolic diameter of greater than 1 in the apical four-chamber view), right-ventricular end-diastolic diameter greater than 30 mm, or paradoxical septal motion. The absence of right-ventricular dysfunction indicates a low likelihood of obstructive shock secondary to PE as the cause of hemodynamic compromise and an alternate source should be identified. The use of the natriuretic peptides and troponin values has a more defined role in the risk stratification and prognostication of patients with an established diagnosis of PE than in initial diagnostic evaluation.

Risk Stratification

Most of the risk stratification tools apply more to the patients on their initial admission and evaluation, and not specifically to ventilated patients who have been in the hospital for some time.

Early mortality rates from PE range from 5% in patients who are hemodynamically stable, to 58% in patients with hemodynamic collapse.^{239,262,263} Clinical variables include prognostication scores in the Pulmonary Embolism Severity Index (PESI)^{264,265} and Simplified PESI²⁶⁶ and were designed to identify patients with low risk for death who could be managed safely as outpatients or candidates for early hospital discharge.

More pertinent to ICU population is the use of troponin, natriuretic peptide and right-ventricle-to-left-ventricle (RV/LV) ratio in identifying patients at risk for adverse outcomes. Previous studies have demonstrated increased short-term mortality associated individually with right-ventricular dysfunction in hemodynamically stable PE ("submassive PE") (defined by RV/LV ratio, pooled from CT angiography and

echocardiographic studies) elevated brain natriuretic peptide (BNP) or N-terminal pro brain natriuretic peptide and myocardial injury as detected by elevated troponin.^{267–273} The PREP study,²⁶³ a prospective cohort study of 570 patients demonstrated combining the individual variables into a prognostic scoring system could further assist in identifying high-risk and low-risk patients. When pooled with the biomarker data, the multivariate model independently associated the highest risk of 30-day adverse events were altered mental status, cardiogenic shock, malignancy, BNP, and RV/LV ratio. The risk factors were used to calculate a prognostic score and based on this score, assigned to three risk classes (low, intermediate, and high risk). In the group as whole, the risk of adverse of events for Class I (low risk), Class II (intermediate risk), and Class III (high risk) were 2.5%, 11.6%, and 43.2%, respectively. In the hemodynamically stable subgroup, the risks were 1.8%, 11.7% and 22.2%, respectively.

Prevention

Prevention of VTE is important in ventilated patients although the optimal method of thromboprophylaxis is difficult to ascertain. The first difficulty is the lack of studies of routine surveillance screening and the lack of a reference standard for the diagnosis of DVT. Ultrasonography is less sensitive as a screening tool, but venography is not uniformly utilized for many reasons. Studies utilizing ultrasound to determine the incidence of DVT in two treatment arms may underestimate the true incidence. Recognizing the variability in the reported studies, chemoprophylaxis, in general, is uniformly superior to mechanical prophylaxis in all critically ill populations, with the exception of neurosurgical patients (in whom chemoprophylaxis is routinely avoided, despite a lack of sound evidence of increased risk of bleeding).^{240,241,247} Recent evidence-based guidelines have been published by the American College of Chest Physicians and by the International Union of Angiology. Despite accumulating evidence of significant reductions in the incidence and mortality of DVT and PE with the use of thromboprophylaxis, and at least three consensus statements guiding appropriate thromboprophylaxis,^{248,274,275} three recent reports point out that only 40% to 60% of at-risk medical patients and 53% to 86% of surgical patients received American College of Chest Physicians recommended prophylaxis.^{276–278}

Studies demonstrate that appropriate chemoprophylaxis with unfractionated heparin, low-molecular-weight heparin, or direct antithrombin III inhibitors (fondaparinux) can reduce the incidence of VTE by approximately 50%, without a significant increase in bleeding.^{240,247,248,274,279–286} The recommendations listed below are from the 2008 *American College of Physicians Clinical Guidelines*. For patients with a moderate risk (medical ICU or postoperative patients), low-molecular-weight heparin or low-dose unfractionated heparin should be used for thromboprophylaxis. For high-risk (trauma or orthopedic surgery) patients, low-molecular-weight heparin should be given. For critical care patients at high risk for bleeding, optimal use of mechanical thromboprophylaxis with graduated compression stockings and/or intermittent pneumatic compression devices should be prescribed at least until the bleeding risk decreases. At that point, pharmacologic thromboprophylaxis should be substituted for or added to the mechanoprophylaxis.

Summary and Conclusions

Critically ill patients are at risk of succumbing to their primary disease or the undesired sequelae associated with their therapy. Although frequently lifesaving, the use of positive airway pressure therapy can have deleterious physiologic consequences. Alterations in renal, hepatic, and gastrointestinal functions have been described. Many of these derangements appear to be a direct result of increased intrathoracic pressure, sympathetic stimulation, and neurohormonal changes. Other indirect complications pertain to therapeutic interventions designed to optimize gas exchange and to alleviate the discomfort of mechanical ventilation. Among these complications are neuromuscular dysfunction, malnutrition, and venous thromboembolism. As ventilators become more complex and offer more options, the potential of unintended consequences increase in tandem. While preventing these untoward events may not be always feasible, recognizing their manifestations may be lifesaving.

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